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A state of the art spectral strategy to a nonlinear dynamical model of HIV/AIDS epidemic

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Abstract. In this paper, a non-linear fractional dynamical system, in our case, modeling HIV/AIDS epidemic is considered. We propose a numerical method based on spectral Galerkin method that handling to the non-smooth behavior of the solution. The most outstanding feature of our approach is to evaluate the approximate solution by means of recurrence relations despite solving complex non-linear algebraic system. At the end, we have given a numerical approximation to illustrate the validity of the method.

Keywords: Mathematical model of HIV/AIDS, Non-linear fractional dynamical system, Fractional Legendre functions, Galerkin method

AMS Mathematics Subject Classification [2010]: 34A09, 65L05, 65L20

1. Introduction

Due to the HIV/AIDS virus and related death toll, many studies have been recorded on mathematical modeling of this epidemic [2]. In [2], authors considered the following model with treatment. The total population is divided into a susceptible class of size z_1, an infectious class before the onset of AIDS and a full-blown AIDS group of size z_4 which is removed from the active population. The infection population is classified into two groups, the asymptomatic stage of size z_2 and the symptomatic stage of size z_3. The fractional-order system is as follows:

$$
\begin{cases}
D_C^{\gamma} z_1(t) = \hat{\beta}\hat{k} - \hat{c}\hat{\nu}(z_2(t) + \hat{a}z_3(t))z_1(t) - \hat{\beta}z_1(t), \\
D_C^{\gamma} z_2(t) = \hat{c}\hat{\nu}(z_2(t) + \hat{a}z_3(t))z_1(t) - (\hat{\beta} + \hat{k}_1)z_2(t) + \hat{\delta}z_3(t), \\
D_C^{\gamma} z_3(t) = \hat{k}_1 z_2(t) - (\hat{\beta} + \hat{k}_2 + \hat{\delta})z_3(t), \\
D_C^{\gamma} z_4(t) = \hat{k}_2 z_3(t) - (\hat{\beta} + \hat{d})z_4(t), \\
z_j(0) = z_j^{(0)}, \quad j = 1,2,3,4, \quad t \in \chi = [0,T],
\end{cases}
\tag{1}
$$

The details of the parameters written in the model (1) and their values are given in Table 1. Here $\gamma = \frac{\eta}{\lambda} \in (0,1)$ is a positive rational number described by the co-prime integers $\eta \geq 1$ and $\lambda \geq 2$, and T is a finite positive real number. D_C^{γ} is known as Caputo fractional derivative of order γ [1].

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In this paper, we intend to provide a highly accurate fractional Legendre Galerkin method for solving the non-linear fractional dynamical system (1).

2. Fractional Legendre Galerkin method

From the well-known existence and uniqueness theorems, we can find that the solutions of (1) suffer from singularity at origin and behave like as $z_j(t) = O(t^\gamma)$, $j = 1, 2, 3, 4$. The main reason of selecting fractional Legendre functions as basis functions is the consistency between the asymptotic behavior of the exact solution, and the degree of smoothness of the basis functions.

The fractional Legendre functions $L_n^{(\tau)}(t)$, $\tau \in (0, 1]$ and $t \in \chi$ are defined from the Legendre polynomials $L_n(x)$ through the coordinate transform $x = 2(\frac{t}{T})^\tau - 1$ as follows [3]

$$L_n^{(\tau)}(t) = L_n\left(2(\frac{t}{T})^\tau - 1\right) = \mathrm{Span}\{1, t^\tau, \ldots, t^{n\tau}\}.$$

These functions are mutually orthogonal concerning the weight function $w^{(\mu,\nu,\tau)}(t) = t^{\tau-1}$. Further properties of fractional Legendre functions can be found in [3]. We set $\tau = \frac{1}{\lambda}$ and for $j = 1, 2, 3, 4$ consider the fractional Legendre Galerkin solution as

$$z_{j,N}(t) = \sum_{i=0}^{\infty} v_{j,i} L_i^{(\tau)}(t) = \underline{v}_j \underline{L} = \underline{v}_j L \underline{T}_t, \quad \underline{v}_j = [v_{j,0}, v_{j,1}, \ldots, v_{j,N}, 0, \ldots], \tag{2}$$

where

$$\underline{L} = [L_0^{(\tau)}(t), L_1^{(\tau)}(t), \ldots, L_N^{(\tau)}(t), \ldots]^T,$$

is the vector of fractional Jacobi functions, L is an infinite lower-triangular matrix and $\underline{T}_t = [1, t^\tau, \ldots, t^{N\tau}, \ldots]^T$.

Now, we give the following lemma which transforms $z_{1,N}(t)z_{2,N}(t)$ and $z_{1,N}(t)z_{3,N}(t)$ into a suitable matrix form.

Lemma 2.1. *The following relations hold*

$$z_{1,N} z_{r,N} = \underline{v}_1 L \mathcal{M}_r \underline{T}_t, \quad r = 2, 3, \tag{3}$$

where $\mathcal{M}_r$ is the following infinite upper-triangular matrices

$$\mathcal{M}_r = \begin{bmatrix} \underline{v}_r L_0 & \underline{v}_r L_1 & \underline{v}_r L_2 & \ldots \\ 0 & \underline{v}_r L_0 & \underline{v}_r L_1 & \ldots \\ 0 & 0 & \underline{v}_r L_0 & \ldots \\ \vdots & \vdots & \vdots & \ddots \end{bmatrix},$$

where L_s,$s = 0, 1, \ldots$ denotes the s-th column of the matrix L.

Since we intend to take the approach of solving the equivalence system of Volterra integral equations of (1), Computing $I^\gamma \underline{T}_t$ is required. Therefore, employing the relations (2), (3) and some simple manipulations yield

$$\begin{cases} \underline{v}_1 L \underline{T}_t = -\hat{c}\hat{\nu}\underline{v}_1 L \mathcal{M}_2 A \underline{T}_t - \hat{c}\hat{\nu}\hat{a}\underline{v}_1 L \mathcal{M}_3 A \underline{T}_t - \hat{\beta}\underline{v}_1 L A \underline{T}_t + \underline{f}_1 \underline{T}_t, \\ \underline{v}_2 L \underline{T}_t = \hat{c}\hat{\nu}\underline{v}_1 L \mathcal{M}_2 A \underline{T}_t + \hat{c}\hat{\nu}\hat{a}\underline{v}_1 L \mathcal{M}_3 A \underline{T}_t - (\hat{\beta} + \hat{k}_1)\underline{v}_2 L A \underline{T}_t + \hat{\delta}\underline{v}_3 L A \underline{T}_t + \underline{f}_2 \underline{T}_t, \\ \underline{v}_3 L \underline{T}_t = \hat{k}_1 \underline{v}_2 L A \underline{T}_t - (\hat{\beta} + \hat{k}_2 + \hat{\delta})\underline{v}_3 L A \underline{T}_t + \underline{f}_3 \underline{T}_t, \\ \underline{v}_4 L \underline{T}_t = \hat{k}_2 \underline{v}_3 L A \underline{T}_t - (\hat{\beta} + \hat{d})\underline{v}_4 L A \underline{T}_t + \underline{f}_4 \underline{T}_t. \end{cases} \tag{4}$$

Here $\underline{f}_j = [f_{j,0}, \ldots, f_{j,N}, \ldots]$, $j = 1, 2, 3, 4$ and A are as follows

$$A = \begin{bmatrix} 0 & \frac{1}{\Gamma(\gamma+1)} & 0 & \cdots & \\ \vdots & 0 & \frac{\Gamma(\tau+1)}{\Gamma(\gamma+\tau+1)} & 0 & \cdots \\ \cdots & \cdots & & \ddots & \ddots \end{bmatrix}, \quad \underline{f}_1 = [\hat{\beta}\hat{k}, 0, \ldots]A + [z_1^{(0)}, 0, \ldots],$$

$$\underline{f}_j = [z_j^{(0)}, 0, \ldots], \; j = 2, 3, 4.$$

It is noticed that I^γ is the Riemann-Liouville fractional integral operator of order γ [1]. Now, Projecting (4) onto $\langle L_0^{(\tau)}(t), ..., L_N^{(\tau)}(t)\rangle$ and defining $\underline{\underline{v}}_j = \underline{v}_j L = [\underline{v}_{j,0}, \underline{v}_{j,1}, \ldots, \underline{v}_{j,N}]$, we arrive at the following system of $4(N+1)$ non-linear algebraic equations

$$\begin{cases} \underline{\underline{v}}_1 = -\hat{c}\hat{\nu}\underline{\underline{v}}_1\mathcal{M}_2^N A^N - \hat{c}\hat{\nu}\hat{a}\underline{\underline{v}}_1\mathcal{M}_3^N A^N - \hat{\beta}\underline{\underline{v}}_1 A^N + \underline{f}_1^N, \\ \underline{\underline{v}}_2 = \hat{c}\hat{\nu}\underline{\underline{v}}_1\mathcal{M}_2^N A^N + \hat{c}\hat{\nu}\hat{a}\underline{\underline{v}}_1\mathcal{M}_3^N A^N - (\hat{\beta} + \hat{k}_1)\underline{\underline{v}}_2 A^N + \hat{\delta}\underline{\underline{v}}_2 A^N + \underline{f}_2^N, \\ \underline{\underline{v}}_3 = \hat{k}_1\underline{\underline{v}}_2 A^N - (\hat{\beta} + \hat{k}_2 + \hat{\delta})\underline{\underline{v}}_3 A^N + \underline{f}_3^N, \\ \underline{\underline{v}}_4 = \hat{k}_2\underline{\underline{v}}_3 A^N - (\hat{\beta} + \hat{d})\underline{\underline{v}}_4 A^N + \underline{f}_4^N, \end{cases} \tag{5}$$

where the index N at the top of the vectors and matrices dictates the principle sub-vectors and sub-matrices of order $N+1$ respectively. Due to the upper-triangular Toeplitz structure of matrices $\mathcal{M}_r^N$, $r = 2, 3$ and the structure of matrix A^N, the unknown components of the unknown vectors $\underline{\underline{v}}_j$, $j = 1, 2, 3, 4$, are derived through the following recurrence relations

$$\begin{cases} \underline{v}_{1,0} = f_{1,0}, \quad \underline{v}_{2,0} = f_{2,0}, \quad \underline{v}_{3,0} = f_{3,0}, \quad \underline{v}_{4,0} = f_{4,0}, \\ \underline{v}_{1,i} = -\hat{c}\hat{\nu}G_{1,2} - \hat{c}\hat{\nu}\hat{a}G_{1,3} - \hat{\beta}F_1 + f_{1,i}, \\ \underline{v}_{2,i} = \hat{c}\hat{\nu}G_{1,2} + \hat{c}\hat{\nu}\hat{a}G_{1,3} - (\hat{\beta} + \hat{k}_1)F_2 + \hat{\delta}F_3 + f_{2,i}, \\ \underline{v}_{3,i} = \hat{k}_1 F_2 - (\hat{\beta} + \hat{k}_2 + \hat{\delta})F_3 + f_{3,i}, \\ \underline{v}_{4,i} = \hat{k}_2 F_3 - (\hat{\beta} + \hat{d})F_4 + f_{4,i}, \quad i = 1, 2, \ldots, N, \end{cases}$$

in which $G_{1,r}$, $r = 2, 3$ are non-linear functions of the elements

$$\underline{v}_{1,0}, \; \underline{v}_{1,1}, \ldots, \; \underline{v}_{1,N}, \; \underline{v}_{r,0}, \; \underline{v}_{r,1}, \ldots, \; \underline{v}_{r,N}.$$

Also, F_s, $s = 1, 2, 3, 4$ are linear functions of the elements $\underline{v}_{s,0}$, $\underline{v}_{s,1}, \ldots,$ $\underline{v}_{s,N}$. Consequently, obtaining $\underline{\underline{v}}_j$, $j = 1, 2, 3, 4$, the approximate solutions (2) can be characterized by solving $\underline{\underline{v}}_j = \underline{v}_j L$. Indeed, our robust and intelligent implementation let us evaluate the unknowns by some recurrence relations without the need to solve non-linear block algebraic system (5) directly.

3. Numerical results

Defining the estimated error

$$\mathbf{E}(N) = \max_{j\in\{1,2,3,4\}} \|e_{j,N}\|_{w^{(\tau)}}, \quad e_{j,N} = z_{j,2N}(t) - z_{j,N}(t),$$

we approximate the solutions of (1) by setting $z_1(0) = 300$, $z_2(0) = 60$, $z_3(0) = 30$, $z_4(0) = 5$, $\hat{k} = 120$, $\hat{a} = 0.3$, $T = 5$, and $\gamma = \frac{4}{5}$. The numerical results are illustrated in Table 2 and Figure 1.

TABLE 1. Description of the Parameters of model (1).

Parameter	*Description*	*Value [?]*
$\hat{\nu}$	*Probability of disease transmission per contact by an infective in stage I*	0.0005
$\hat{k}_1$	*Transfer rates from stage I to stage J*	0.01
$\hat{k}_2$	*Transfer rates from stage J to AIDS stage*	0.02
$\hat{\beta}$	*The natural death rate*	0.02
$\hat{d}$	*The death rate for AIDS*	0.075
$\hat{c}$	*The average number of contacts that an individual has per unit time*	3
$\hat{\delta}$	*The treatment rate from stage J to stage I*	0.01

TABLE 2. The numerical results for various values of N.

N	$\mathbf{E}(N)$	*Convergence order*	*CPU-time*
60	1.49×10^{-1}	-	3.25
120	1.01×10^{-2}	3.88	15.27
240	8.95×10^{-6}	10.14	95.53
480	2.57×10^{-12}	21.73	1716.27

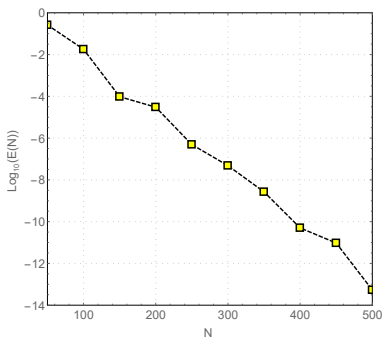

FIGURE 1. Illustration for semi-log errors.

4. Conclusion

In this paper, a suitable Galerkin approach based on the fractional Legendre functions was implemented to approximate the solutions of a non-linear dynamical model of HIV/AIDS epidemic. This strategy offered some recurrence relations for deriving the approximate solution rather than solving a non-linear complex algebraic system.

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